

Sandeep Kambhampati



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Education

Harvard University (Boston, MA)

Expected December 2026

Ph.D. candidate, Bioinformatics and Integrative Genomics

Johns Hopkins University (JHU) (Baltimore, MD)

May 2021

B.S., Biomedical Engineering (BME) | Genomics & Systems Biology Track | GPA: 3.93

Technical Skills

Proficient: Python, PyTorch, Lightning, Pyro, R, Unix, Git

Basic: JAX/NumPyro, Julia, Java

Experienced in Self-Supervised Learning, Generative models, Computer Vision, Distributed Training, Model Optimization, Bayesian Inference, Generalized Linear Models

Research Experience

Microsoft Research New England – *Research Intern*

Summer 2025

Advisor: Dr. Alex Lu (*Senior Researcher*)

- Designed *Vermeer*, a channel-adaptive autoregressive transformer that generates microscopy images of protein localization and generalizes to previously unseen proteins
- Applying *Vermeer* as a *virtual staining* model to investigate how pathogenic mutations alter protein localization and overcome limitations in experimental data collection

Broad Institute of MIT & Harvard – *Graduate Research Student*

2021-present

Advisors: Drs. Fei Chen (*Core Member*), Mehrtash Babadi (*Director of Computational Methods*)

- Built *TissueMosaic*, a self-supervised representation learning framework for spatial transcriptomics (ST) data, by adapting DINO with domain-specific augmentations.
- Formulated a latent neural PDE framework to model how interactions between local spatial processes give rise to developmental patterns in time-series ST data
- Developing a Point Cloud Transformer foundation model to learn rules of tissue organization

Johns Hopkins University – *Undergraduate Researcher*

2017-2021

- Constructed a statistical physics model of DNA methylation from long read ONT sequencing data and applied this model to study genetic heritability of DNA methylation patterns
- Analyzed RNA-seq data to identify key transcriptional regulators of multi-organ development
- Engineered mouse knockout and *in vitro* overexpression models to identify gene targets that enhance the therapeutic potential of stem-cell derived cardiomyocytes

Publications (selected)

1. **Kambhampati, S.**, ... Chen, F.#, Lu, A.# Vermeer: Autoregressive generative modeling of microscopy predicts protein localization. (2026) [Github](#) [bioRxiv](#)
2. Davidovich, A., ... **Kambhampati, S.**, *et al.* Non-Mendelian inheritance of DNA methylation patterns in mice. *Nature Genetics* (2026)
3. Bhate, S.*, **Kambhampati, S.***, Babadi, M., Chen, F.#, Uhler, C.# DALI Learns Rules Generating Spatiotemporal Transcriptomics. *ICLR GEN2 Workshop.* (2026) [Github](#) [Paper](#)

4. **Kambhampati, S.**, D'Alessio, L., Grab, F., Fleming, S., Chen, F.#, Babadi, M.# *TissueMosaic* enables cross-sample differential analysis of spatial transcriptomics datasets through self-supervised representation learning. **Cell Systems** (2025) [Github Paper](#)
5. Russell, A.*, Weir, J.*, Nadaf, N.* ... **Kambhampati, S.**, *et al.* Slide-tags enables single-nucleus barcoding for multimodal spatial genomics. **Nature** (2023)
6. **Kambhampati, S.***, Murphy, S.*, *et al.* Cross-organ transcriptomic comparison reveals universal factors during maturation. **Journal of Computational Biology** (2022)
7. Abanti, J., **Kambhampati, S.**, Feinberg, A., Goutsias, J. Estimating DNA methylation potential energy landscapes from nanopore sequencing data. **Scientific Reports** (2021)
8. Murphy, S., ... **Kambhampati, S.**, *et al.* PGC1/PPAR Drive Cardiomyocyte Maturation through Regulation of Yap1 and SF3B2. **Nature Communications** (2021)

* or # denotes equal contribution

Oral Presentations (selected)

Boston, USA	Eric and Wendy Schmidt Center Symposium: Biomedical Science/AI	2026
Seoul, South Korea	Research in Computational Molecular Biology (RECOMB)	2025
Vancouver, Canada	Keystone Symposia: Single-cell Biology and Tissue Genomics	2024

Leadership and Outreach

Health Professions Recruitment Program – Mentor 2021 – 2023

- Mentored Boston high schoolers from underserved backgrounds to produce a research project and form a comprehensive plan to achieve their goals in science and medicine

Hopkins Undergraduate Research Journal – Editor-in-Chief 2018 – 2021

- Published an annual journal of 15+ STEM and Humanities articles
- Established a faculty review process and a mission on accessible scientific communication

Thread – Head of Family 2017 - 2021

- Managed fellow volunteers to provide a support network “family” for Baltimore high school students confronting significant barriers outside of the classroom

Teaching and Mentorship

Eric Wang (MIT undergraduate)	2026
Qiyu Gong (Broad Institute, Associate Computational Biologist)	2022-2025
Ebtisam Alshehri (Broad Institute, Machine Learning Research Intern)	2024
Matthew Shum (Harvard undergraduate)	2022
Teaching Assistant: Systems and Controls (JHU BME Core Curriculum)	2019, 2020

Awards (selected)

NSF Graduate Research Fellowship Program (GRFP) Covers PhD tuition+stipend for 3 years	2022
Hertz Fellowship Finalist 1 of 45 finalists selected from 650+ applicants	2022
Tau Beta Pi Honor Society Top 1/5 of JHU seniors in engineering	2021
Frances Howard Flatau Scholarship To 1 JHU BME upperclassman annually	2020, 2021

Hobbies and Interests

Rock climbing, Harvard Mountaineering Club, sailing, mountain biking, chess, Indian classical violin